pandas.DataFrame.rename

copy=None, inplace=False, level=None, errors='ignore') [source]

Rename columns or index labels.

Function / dict values must be unique (1-to-1). Labels not contained in a dict / Series will be left as-is. Extra labels listed don't throw an error.

See the user guide for more.

Parameters:

mapper : *dict-like or function*

Dict-like or function transformations to apply to that axis' values. Use either `mapper` and `axis` to specify the axis to target with `mapper`, or `index` and `columns`.

index : *dict-like or function*

Alternative to specifying axis (`mapper, axis=0` is equivalent to `index=mapper`).

columns : *dict-like or function*

Alternative to specifying axis (`mapper, axis=1` is equivalent to `columns=mapper`).

axis : *{0 or 'index', 1 or 'columns'}, default 0*

Axis to target with `mapper`. Can be either the axis name ('index', 'columns') or number (0, 1). The default is 'index'.

copy : *bool, default True*

Also copy underlying data.

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Note

The *copy* keyword will change behavior in pandas 3.0. Copy-on-Write will be enabled by default, which means that all methods with a *copy* keyword will use a lazy copy mechanism to defer the copy and ignore the *copy* keyword. The *copy* keyword will be removed in a future version of pandas.
You can already get the future behavior and improvements through enabling copy on write `pd.options.mode.copy_on_write = True`

inplace : *bool, default False*

Whether to modify the DataFrame rather than creating a new one. If True then value of copy is ignored.

level : *int or level name, default None*

In case of a MultiIndex, only rename labels in the specified level.

errors : *{'ignore', 'raise'}, default 'ignore'*

If 'raise', raise a *KeyError* when a dict-like *mapper*, *index*, or *columns* contains labels that are not present in the Index being transformed. If 'ignore', existing keys will be renamed and extra keys will be ignored.

Returns:

DataFrame or None

DataFrame with the renamed axis labels or None if `inplace=True`.

Raises:

KeyError

If any of the labels is not found in the selected axis and "errors='raise'".

See also

`DataFrame.rename_axis`
Set the name of the axis.

Examples

`DataFrame.rename` supports two calling conventions

- `(index=index_mapper, columns=columns_mapper, ...)`

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We *highly* recommend using keyword arguments to clarify your intent.

Rename columns using a mapping:

```
>>> df = pd.DataFrame({"A": [1, 2, 3], "B": [4, 5, 6]})
>>> df.rename(columns={"A": "a", "B": "c"})
   a  c
0  1  4
1  2  5
2  3  6
```

Rename index using a mapping:

```
>>> df.rename(index={0: "x", 1: "y", 2: "z"})
   A  B
x  1  4
y  2  5
z  3  6
```

Cast index labels to a different type:

```
>>> df.index
RangeIndex(start=0, stop=3, step=1)
>>> df.rename(index=str).index
Index(['0', '1', '2'], dtype='object')
```

```
>>> df.rename(columns={"A": "a", "B": "b", "C": "c"}, errors="raise")
Traceback (most recent call last):
KeyError: ['C'] not found in axis
```

Using axis-style parameters:

```
>>> df.rename(str.lower, axis='columns')
   a  b
0  1  4
1  2  5
2  3  6
```

```
>>> df.rename({1: 2, 2: 4}, axis='index')
   A  B
0  1  4
2  2  5
```


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